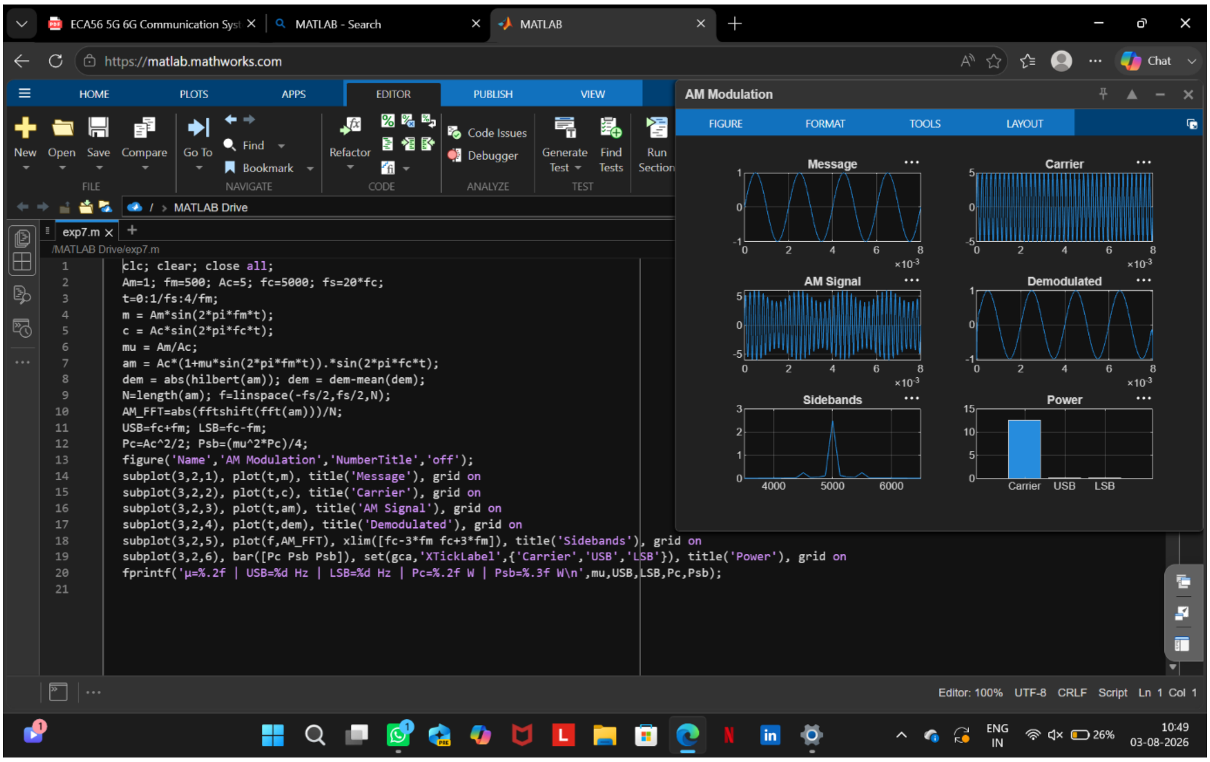
EXP:1 AMPLITUDE MODULATION

1.2 Characteristics study of Side Band power and Frequencies in AM using MATLAB

CODE:

```
clc;
clear;
close all;
Am=1;
fm=500;
 Ac=5;
 fc=5000;
 fs=20*fc;
t=0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
mu = Am/Ac;
am = Ac*(1+mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
dem = abs(hilbert(am)); dem = dem-mean(dem);
N=length(am); f=linspace(-fs/2,fs/2,N);
AM_FFT=abs(fftshift(fft(am)))/N;
USB=fc+fm; LSB=fc-fm;
Pc=Ac^2/2; Psb=(mu^2*Pc)/4;
figure('Name','AM Modulation','NumberTitle','off');
subplot(3,2,1), plot(t,m), title('Message'), grid on
subplot(3,2,2), plot(t,c), title('Carrier'), grid on
subplot(3,2,3), plot(t,am), title('AM Signal'), grid on
subplot(3,2,4), plot(t,dem), title('Demodulated'), grid on
subplot(3,2,5), plot(f,AM_FFT), xlim([fc-3*fm fc+3*fm]), title('Sidebands'), grid on
subplot(3,2,6), bar([Pc Psb Psb]), set(gca,'XTickLabel',{'Carrier','USB','LSB'}),
title('Power'), grid on
fprintf('μ=%.2f | USB=%d Hz | LSB=%d Hz | Pc=%.2f W | Psb=%.3f
W\n',mu,USB,LSB,Pc,Psb);
```

OUTPUT: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New Open Save Compare Go To Find Bookmark Refactor Code Issues Debugger Generate Test Find Tests Run Section
FILE NAVIGATE CODE ANALYZE TEST
MATLAB Drive
exp7.m
/MATLAB Drive/exp7.m
clc; clear; close all;
Am=1; fm=500; Ac=5; fc=5000; fs=20*fc;
t=0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
mu = Am/Ac;
am = Ac*(1+mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
dem = abs(hilbert(am)); dem = dem-mean(dem);
N=length(am); f=linspace(-fs/2,fs/2,N);
AM_FFT=abs(fftshift(fft(am)))/N;
USB=fc+fm; LSB=fc-fm;
Pc=Ac^2/2; Psb=(mu^2*Pc)/4;
figure('Name','AM Modulation','NumberTitle','off');
subplot(3,2,1), plot(t,m), title('Message'), grid on
subplot(3,2,2), plot(t,c), title('Carrier'), grid on
subplot(3,2,3), plot(t,am), title('AM Signal'), grid on
subplot(3,2,4), plot(t,dem), title('Demodulated'), grid on
subplot(3,2,5), plot(f,AM_FFT), xlim([fc-3*fm fc+3*fm]), title('Sidebands'), grid on
subplot(3,2,6), bar([Pc Psb Psb]), set(gca,'XTickLabel',{'Carrier','USB','LSB'}), title('Power'), grid on
fprintf('μ=%.2f | USB=%d Hz | LSB=%d Hz | Pc=%.2f W | Psb=%.3f W\n',mu,USB,LSB,Pc,Psb);
AM Modulation
FIGURE FORMAT TOOLS LAYOUT
Message
Carrier
AM Signal
Demodulated
Sidebands
Power
Carrier USB LSB
Editor: 100% UTF-8 CRLF Script Ln 1 Col 1
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